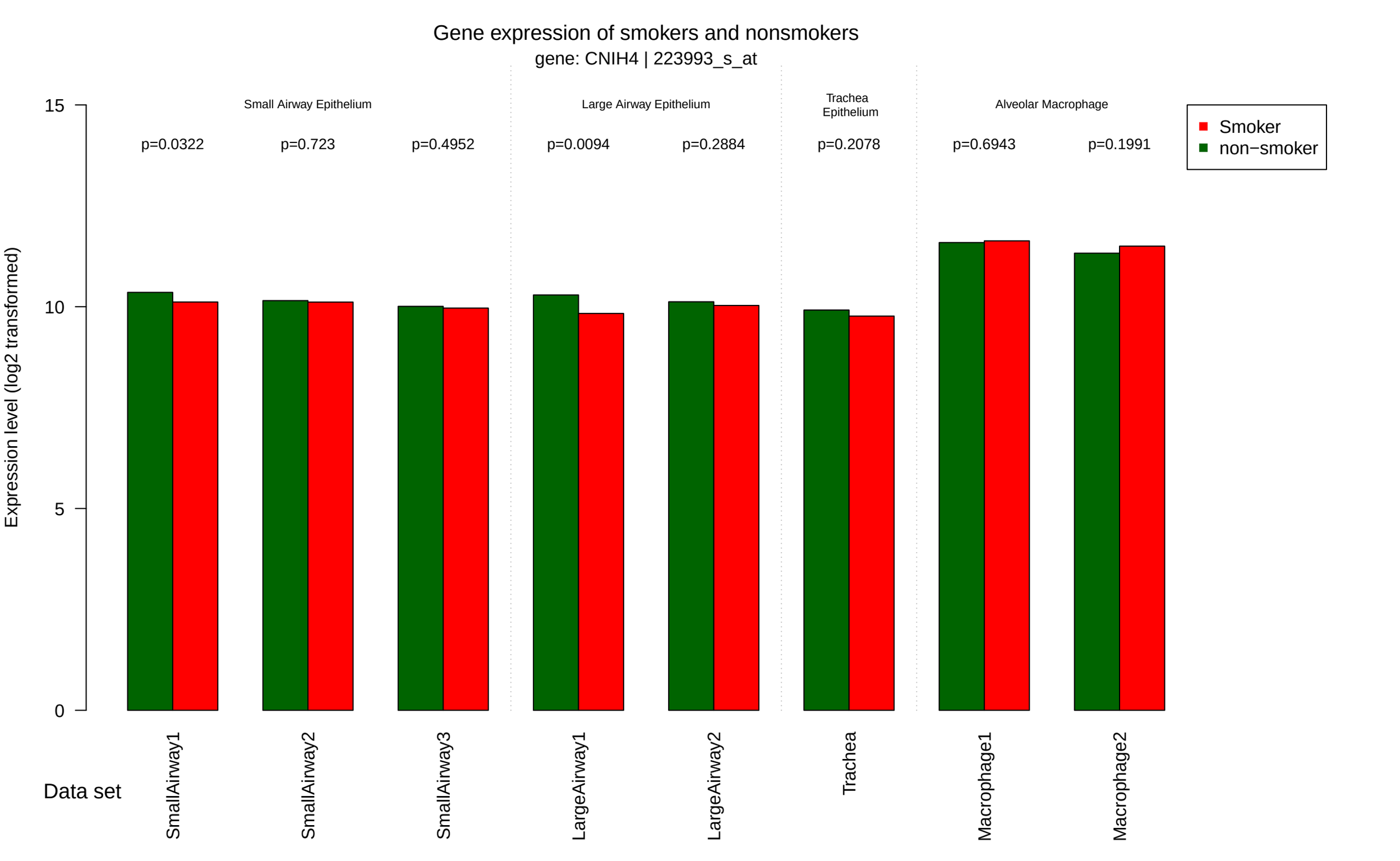
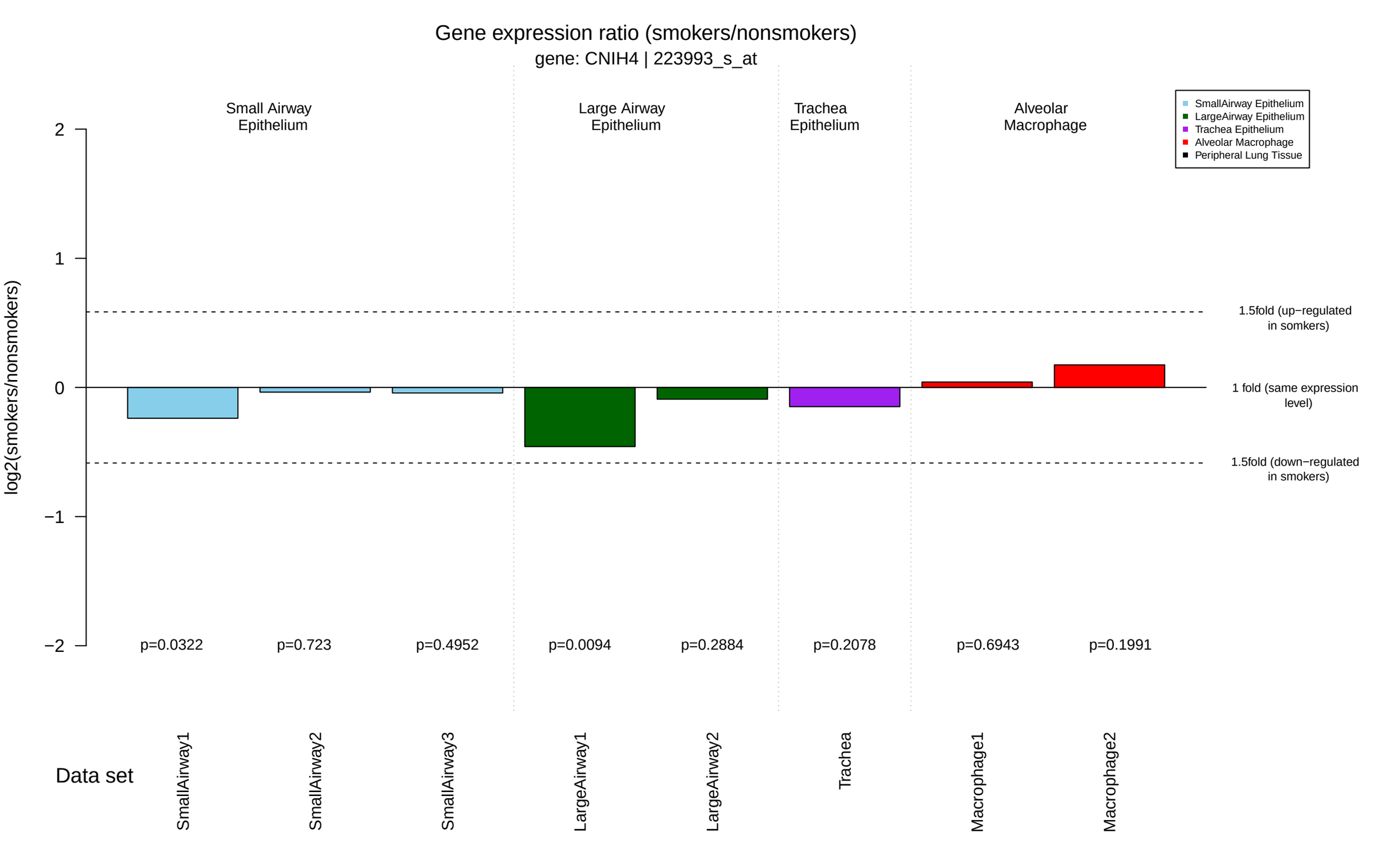
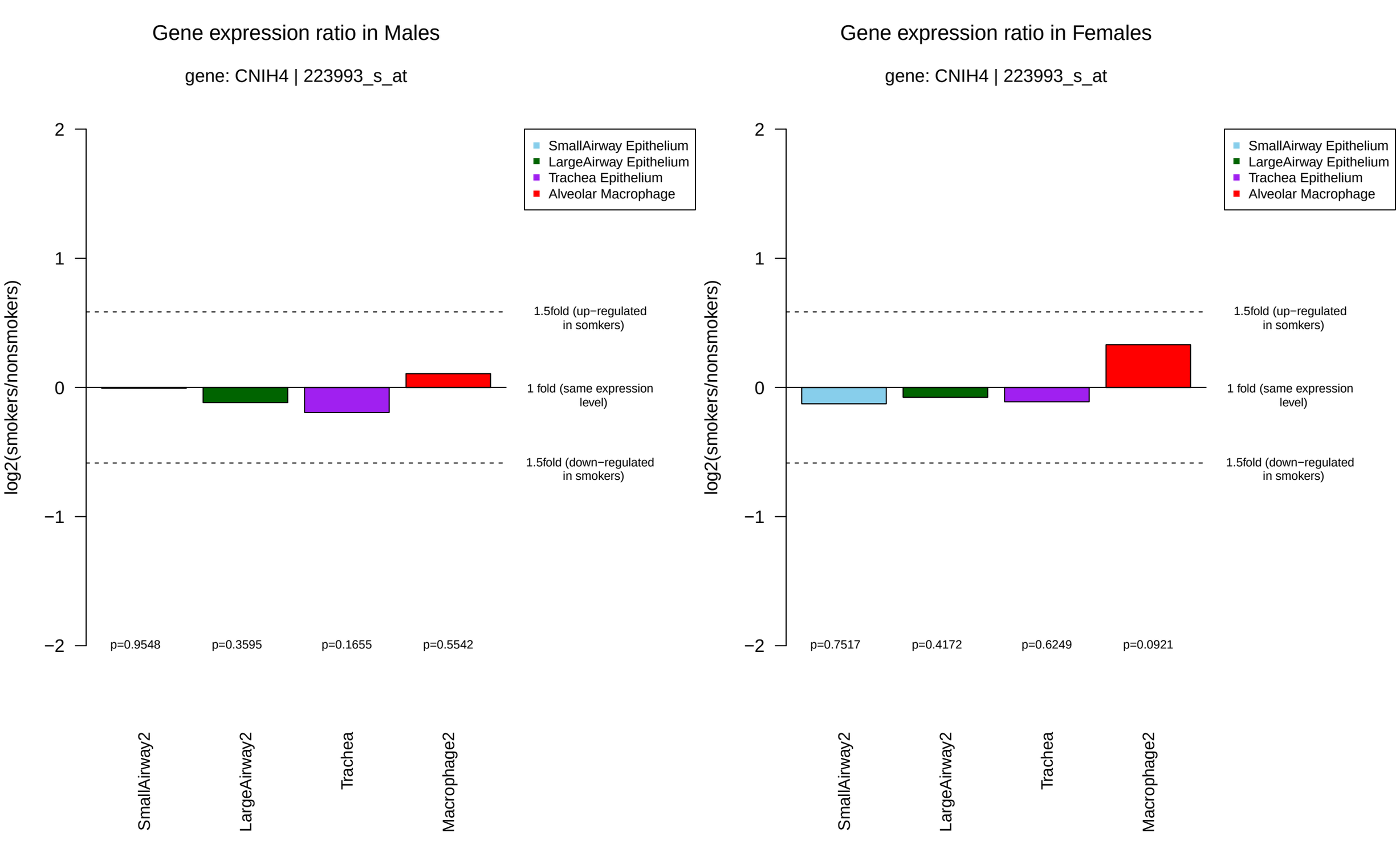
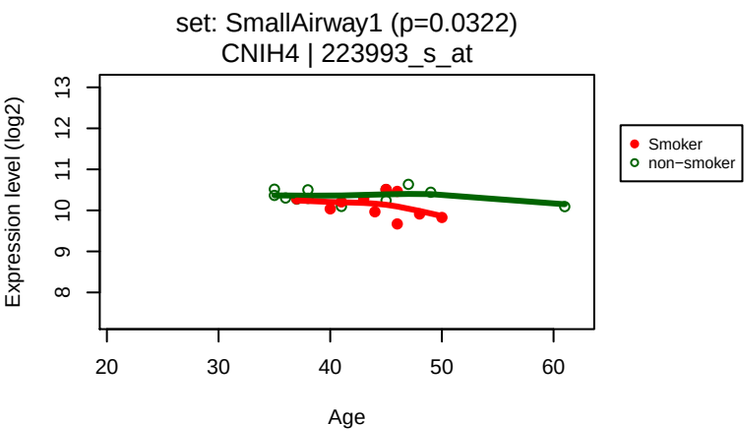
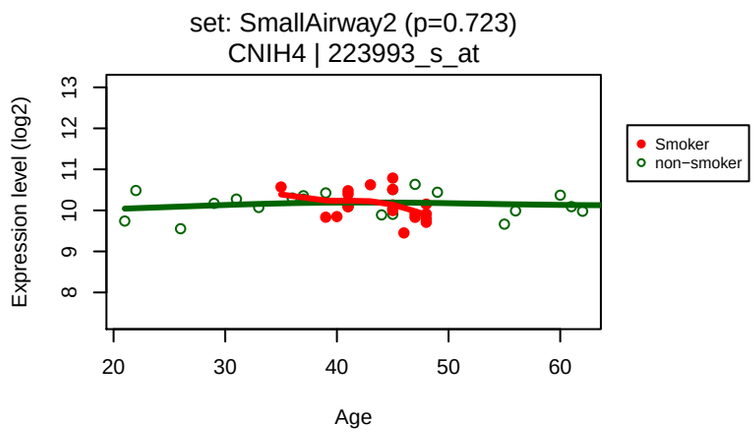
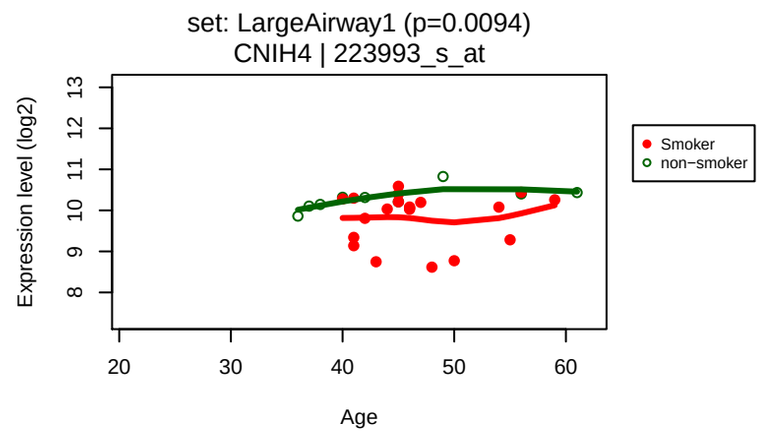

Gene expression of smokers and nonsmokers
gene: CNIH4 | 223993_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0322
p=0.723
p=0.4952
p=0.0094
p=0.2884
p=0.2078
p=0.6943
p=0.1991
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

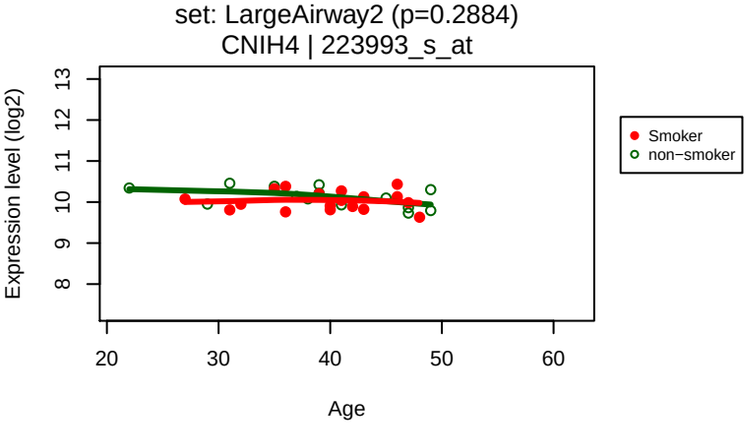

Gene expression ratio (smokers/nonsmokers)
gene: CNIH4 | 223993_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0322
p=0.723
p=0.4952
p=0.0094
p=0.2884
p=0.2078
p=0.6943
p=0.1991
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

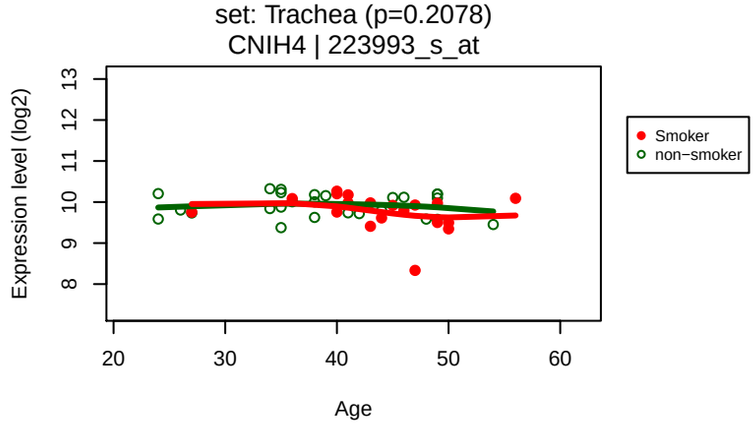

Gene expression ratio in Males
gene: CNIH4 | 223993_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9548
p=0.3595
p=0.1655
p=0.5542
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CNIH4 | 223993_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7517
p=0.4172
p=0.6249
p=0.0921
SmallAirway2
LargeAirway2
Trachea
Macrophage2

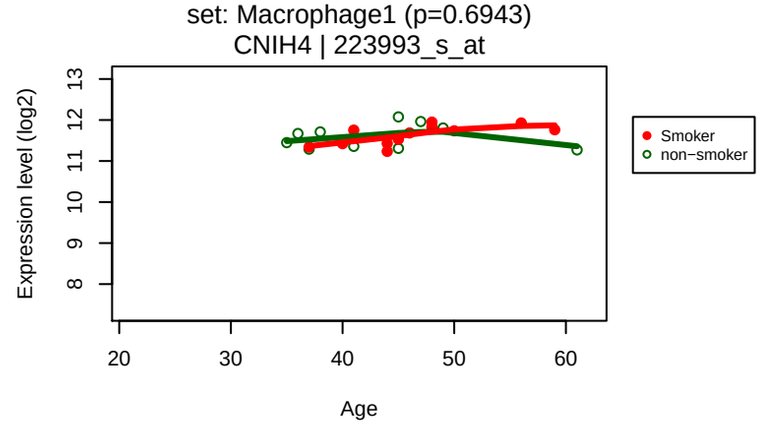

set: SmallAirway1 (p=0.0322)
CNIH4 | 223993_s_at
Expression level (log2)
Age
Smoker
non−smoker

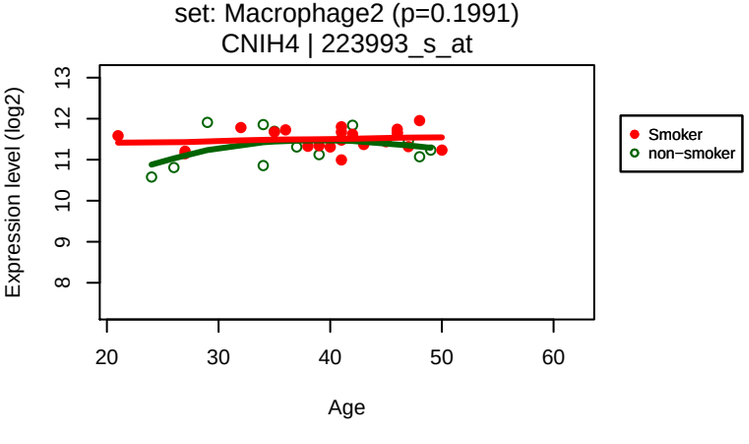

set: SmallAirway2 (p=0.723)
CNIH4 | 223993_s_at
Expression level (log2)
Age
Smoker
non−smoker

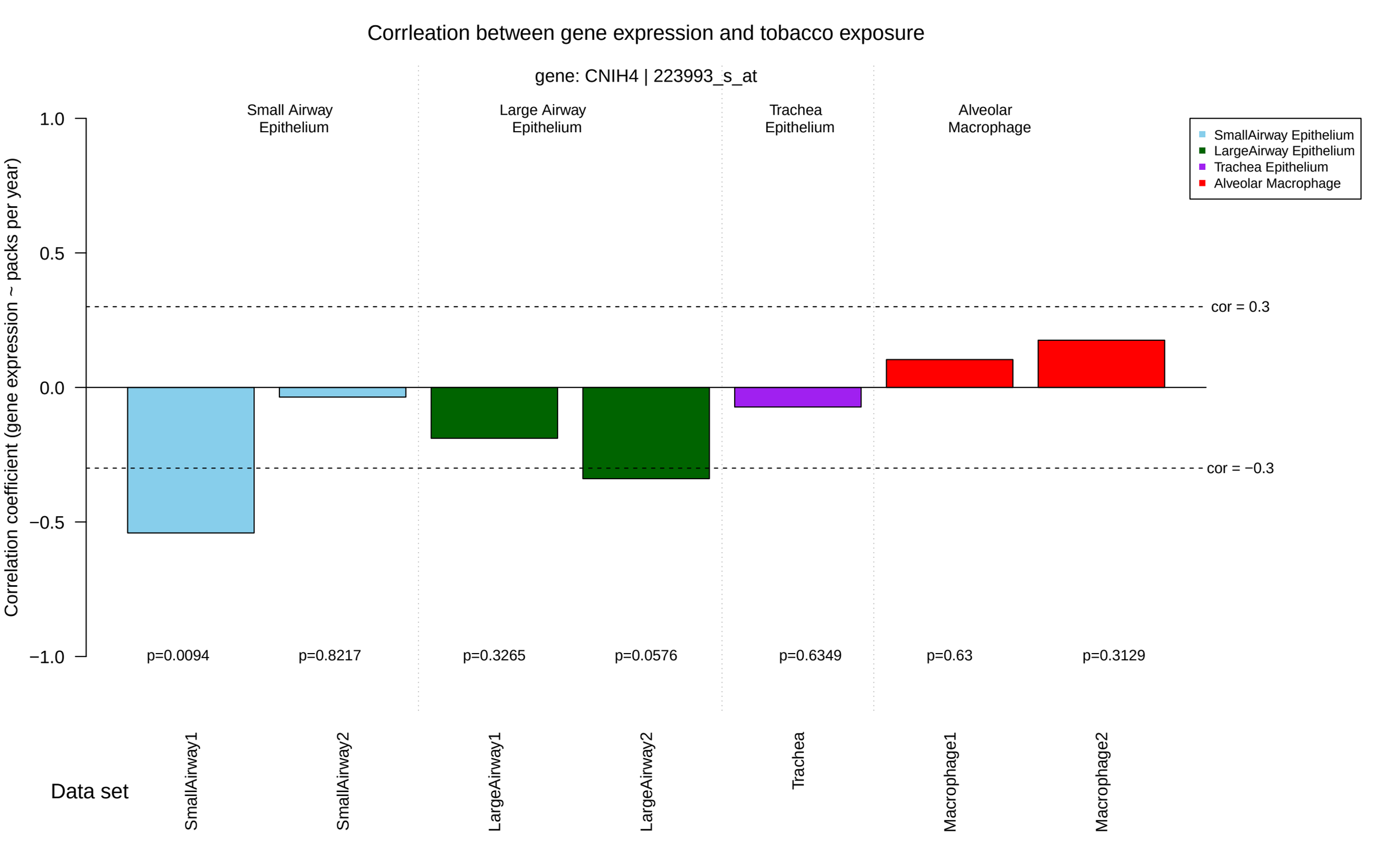

set: LargeAirway1 (p=0.0094)
CNIH4 | 223993_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2884)
CNIH4 | 223993_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2078)
CNIH4 | 223993_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6943)
CNIH4 | 223993_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1991)
CNIH4 | 223993_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CNIH4 | 223993_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0094
p=0.8217
p=0.3265
p=0.0576
p=0.6349
p=0.63
p=0.3129
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2